

Supplementary Online Materials to:

Data and Code Availability in Sports Science: A Registered Report

Registered Report – Stage 1

PREPRINT VERSION – NOT PEER REVIEWED

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19 S1: Article Search Strings

20

21 Search string for Web of Science:

22

23 “(SO=(JOURNAL1 OR JOURNAL2 OR JOURNALK)) AND PY=(2017-2026) AND DT=(Article)”

24 where JOURNALK is a placeholder for the k journals that match the inclusion criteria.

25

S2: Pilot Analysis

Aim:

A pilot analysis was conducted to test the PDF retrieval and automated extraction tools. We randomly sampled 200 articles from a population sports science that will not be part of our study dataset. We documented the success rate and time required for automated and manual PDF retrieval and data extraction. We compared the results of the manual and automated extraction to determine the error rates of the automated procedure and to inform the extraction procedures for the main project.

Methods:

We screened the SCImago Journal Rank 2025 (SCImago, 2025) in the subject category “Sports Science” for journals that were not listed as Q1 in the category during the previous ten years. We selected the ten journals with the highest 2025 rankings after excluding three journals that primarily publish non-quantitative research (International Review for the Sociology of Sport, Sociology of Sport Journal, Journal of the Philosophy of Sports). The remaining journals were: Journal of Human Kinetics; Strength and Conditioning Journal¹; Journal of Sports Science and Medicine; Physical Therapy in Sport; Physician and Sportsmedicine; Sports Biomechanics; International Journal of Sports Science & Coaching; Journal of Sports Economics; Research Quarterly for Exercise and Sport; International Journal of Sports Marketing & Sponsorship.

We performed an article search using Web of Science on 28 August 2026 based on the inclusion criteria specified in our article. The exact search term was:

“(SO=(Journal of Human Kinetics OR Strength and Conditioning Journal OR Journal of Sports Science and Medicine OR Physical Therapy in Sport OR Physician and Sportsmedicine OR Sports Biomechanics OR International Journal of Sports Science Coaching OR Journal of Sports Economics OR Research Quarterly for Exercise and Sport OR International Journal of Sports Marketing Sponsorship)) AND PY=(2017-2026) AND DT=(Article)”

The search results were exported as .txt files and imported into R, where we took a random sample of n = 200 articles to constitute the pilot sample. We then attempted to retrieve the articles’ PDFs as outlined in the manuscript and documented the success rate for automated retrieval and the time required for manual retrieval.

To validate the automated screening process, we manually extracted all relevant items from the texts. We then conducted the automated screening and compared the results of the two approaches. During reviewing discrepancies between the automated and the manual extraction, we refined the automated extraction procedure and reran it.

Results:

We could retrieve the full texts for n = 195 articles². The Zotero import function for PDF retrieval worked for 53 (26.5%) articles. The average time required for retrieving a PDF manually was 16.5

¹ During the reanalysis we realized that “Strength and Conditioning Journal” mainly publishes non-original research and thus does not meet the inclusion criteria. However, we retained the few articles from this journal (n = 9) in our sample to test whether the automated extraction can correctly identify that these articles do not contain data or code availability statements.

² Five articles were recently published in one journal that was included in our university’s subscription system only for articles assigned to an issue, but not for articles published ahead of print. Because this was a pilot study, we excluded these articles without resampling.

seconds. Manual extraction took an average of 52 seconds per study, while automated extraction took 6 seconds per article.

Data and/or code availability was automatically flagged in 42 cases, of which 29 corresponded to cases identified through manual extraction, 2 identified statements overlooked during manual extraction, and 13 were false positives (13/195, 7%). Note that these false positives would have been identified during the data extraction for the main project, which includes a manual review of each automatically flagged entry. The false negative rate was 0%, i.e., no data/code sharing or availability statement was overlooked by the automated procedure in the pilot sample.

Regarding the analysis software used, a total of 43 discrepancies across 40 articles were found. However, after correcting for errors (overlooked uses of software) in the manual extraction, only 17 discrepancies remained; 3 were false negatives (i.e., the automatic extraction overlooked the use of a software), and 14 were false positive (i.e., the automated extraction incorrectly detected the use of a software). Notably, 12 of the 14 false positives were related to the use of the software *R*, because regular expression matching proved difficult in these cases.

Conclusions:

- Retrieving PDFs is a bottleneck for scalability, as we can assume that 75% of the included studies will require manual retrieval, with approximately 16.5 seconds needed for each download.
- Automated extraction is approximately 10 times faster than manual extraction and has fewer extraction errors for the item analysis software used.
- Automatically flagging data/code availability statements has a false positive rate of around 7%, and thus requires manual oversight. The false negative rate is very low.
- Automatically detecting the analysis software in a study has false-positive and false-negative rates of approximately 1-2% each, provided that studies in which the software *R* is detected are manually reviewed (as these show high false positive rates).

S3: Automated Extraction Strategy

Articles are flagged for manual extraction if one of the following conditions occur:

- A data or code availability statement is likely present:
 - o The grobid import recognizes an availability statements OR
 - o The full article text matches one of the following strings: "data availability" OR "code availability" OR "(on|by|upon)\s+(reasonable\s+)?request" OR "datasets?\s+(?:is|are)?\s*(?:not\s+)?(?:publicly\s+)?available" OR "available\s+from\s+(?:the\s+)?(?:corresponding\s+)?author"
- Data or code is likely shared:
 - o The full article text matches one of the following strings that may indicate a data/code repository: "repositor" OR "archive" OR "\bosf\b" OR "\osf.io" OR "open science framework" OR "researchbox" OR "zenodo" OR "github" OR "figshare" OR "datadryad" OR "kaggle" or "mendeley" OR "ukbiobank" OR "ukdataservice" OR "dataverse" OR "clinicalstudydatarequest" OR "ourworldindata"

Some of the regular expression strings are modified from the code of the *open_practices* module in the *metacheck* R package (DeBruine et al., 2026). Every article that is flagged will be manually reviewed.

For the determination of statistical software used, we use the following search strings:

- SPSS: "\bspss" OR "statistical package for the social sciences"
- Stata: "stata"
- jamovi: "jamovi"
- JASP: "\bjasp\b"
- R: "(?<!\bR(?:=\s(?:!\s*)))" (case-sensitive!) OR "(?:in|software)\s+R\b" (case-sensitive!) OR "a language and environment for statistical computing" OR "rstudio" OR "\br language"
- Python: "python" OR "scipy"
- GraphPad Prism: "graph pad" OR "\bprism\b"
- Excel: "excel\b"
- SAS: "\bsas\b"
- LISREL: "lisrel"
- MATLAB: "matlab"
- Statistica: "statistica"
- WinBUGS: "winbugs"
- BioEstat: "bioestat"
- Stan: "\bstan\b"
- PSPP: "\bpspp"
- MedCalc: "medcalc"

The list of software packages is in part informed by the work of Borg et al. (2020). When an article is flagged for using R, it will be manually reviewed.

The full R code for the above procedures is presented here:

```
#' Flag data availability statements and repository mention
#
#' Detects whether an article contains a data or code availability statement
#' and whether it mentions a data repository. Availability statements are
#' identified either through a dedicated availability section (identified by
# the grobid import) or by searching
```

```

141 #' for common phrases (e.g., "data availability", "code availability", or
142 #' "available upon reasonable request"). Repository mentions are detected by
143 #' searching for references to common repositories such as OSF, Zenodo,
144 #' GitHub, Figshare, Dryad, Dataverse, and others.
145 #'
146 #' @param article An object with class "scivrs_paper", usually imported
147 #'   by the metacheck package
148 #'
149 #' @return A data frame with one row and two logical columns:
150 #' \describe{
151 #'   \item{flag_statement}{`TRUE` if the article contains a data/code
152 #'     availability statement or a dedicated availability section.}
153 #'   \item{flag_repo}{`TRUE` if the article mentions a supported data
154 #'     repository.}
155 #' }
156 #'
157 flag_article <- function(article) {
158   # search for availability section in grobid import
159   if (is.null(article$section)) article <- article[[1]] # get list item if
160   needed
161   sec <- any(article$section$section_type == "availability")
162   # search for strings that indicate availability section
163   st <- sfind(
164     article,
165     c("data availability", "code availability",
166       "(on|by|upon)\\s+(reasonable\\s+)?request",
167       "datasets?\\s+(?:is|are)?\\s*(?:not\\s+)?(?:publicly\\s+)?available",
168       "available\\s+from\\s+(?:the\\s+)?(?:corresponding\\s+)?author"
169     ),
170     search_header = TRUE
171   )
172   # search for strings that match data/code repository
173   repo <- sfind(
174     article,
175     c("repositor", "archive", "\\bosf\\b", "osf\\.io",
176       "open science framework", "researchbox", "zenodo", "github",
177       "figshare", "datadryad", "kaggle", "mendeley", "ukbiobank",
178       "ukdataservice", "dataverse", "clinicalstudydatarequest",
179       "ourworldindata")
180   )
181   data.frame(
182     flag_statement = any(c(sec, st)),
183     flag_repo = repo
184   )
185 }
186
187 #' Detect statistical software mentioned in an article
188 #'
189 #' Searches an article for mentions of commonly used statistical software and

```

```

190 #' programming languages. Detection is based on regular expressions and common
191 #' alternative names where appropriate (e.g., "Statistical Package for the
192 #' Social Sciences" for SPSS or "RStudio" for R).
193 #'
194 #' @param article An object with class "scivrs_paper", usually imported
195 #'   by the metacheck package
196 #'
197 #' @return A data frame with one row containing logical columns indicating
198 #'   whether each software package was detected. The returned columns are:
199 #'   `spss`, `stata`, `jamovi`, `jasp`, `r`, `python`, `prism`, `excel`,
200 #'   `sas`, `lisrel`, `matlab`, `statistica`, `winbugs`, `bioestat`,
201 #'   `stan`, `pspp`, and `medcalc`.
202 #'
203 extract_software <- function(article) {
204   data.frame(
205     spss = sfind(article, c("\\bspss", "statistical package for the social
206 sciences", "\\bpasw", "\\bamos")),
207     stata = sfind(article, "\\bstata"),
208     jamovi = sfind(article, "\\bjamovi"),
209     jasp = sfind(article, "\\bjasp"),
210     r = any(
211       sfind(article, c("(?!-)\bR(?:=\\s(?:!\\s*))",
212 "(?:in|software)\\s+R\\b"), ignore.case = FALSE, perl = TRUE),
213       sfind(article, c("a language and environment for statistical
214 computing", "rstudio", "\\br language"))
215     ),
216     python = sfind(article, c("python", "scipy")),
217     prism = sfind(article, c("graph pad", "\\bprism\\b")),
218     excel = sfind(article, "excel\\b"),
219     sas = sfind(article, "\\bsas\\b"),
220     lisrel = sfind(article, "lisrel"),
221     matlab = sfind(article, c("matlab", "mathworks")),
222     statistica = sfind(article, "\\bstatistica\\b"),
223     winbugs = sfind(article, "winbugs"),
224     bioestat = sfind(article, "bioestat"),
225     stan = sfind(article, "\\bstan\\b"),
226     pspp = sfind(article, "\\bpspp"),
227     medcalc = sfind(article, "medcalc")
228   )
229 }
230 #' Helper function to find matches using metacheck::text_search()
231 #'
232 #' Searches an article text via metacheck::text_search() and returns
233 #' if any matches were found
234 #'
235 #' @param article An object with class "scivrs_paper", usually imported
236 #'   by the metacheck package
237 #' @param string Regular expression pattern passed to `pattern` in
238 #'   metacheck::text_search()

```

```
239 #' @param ... Additional arguments passed to metacheck::text_search()
240 #'
241 #' @return TRUE or FALSE, whether a match was found.
242 sfind <- function(article, string, ...) {
243   t <- metacheck::text_search(paper = article, pattern = string, ...)
244   nrow(t) >= 1
245 }
246
```

247 S4: Study Design Template

Question	Hypothesis	Sampling plan	Analysis Plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes	Theory that could be shown wrong by the outcomes
Has the availability of data (H1) and code (H2) in published sports science articles changed over the last ten years?	The availability of data/code has not changed over the last ten years (H1_0/H2_0). The availability of data/code has changed over the last ten years (H1_1/H2_1).	We will investigate the full population of all original articles published in the last ten years (2017-2026) in all sports science journals that were listed as Q1 in at least three years during this period according to the SCImago Journal Rank.	We will run a logistic model of the form: Shared ~ time + (time journal)	We will investigate the full population of Q1 sports science articles for the last ten years	If the 95%-CI for the main effect of time (odds ratio) excludes 1, we will reject the H_0.	n/a

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References

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